EPFL

Final Project

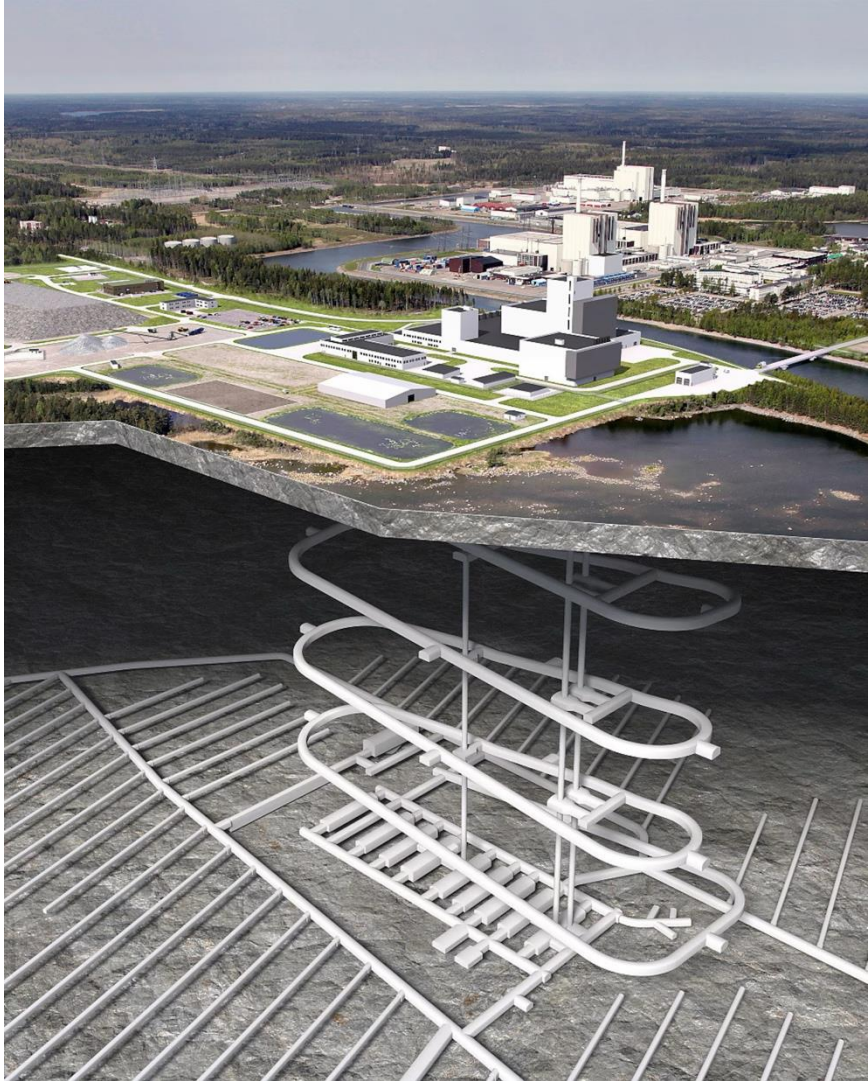
Modeling of a repository tunnel and nuclear waste canister temperatures over time

Mont Terri Project

23.04.2026

Outline

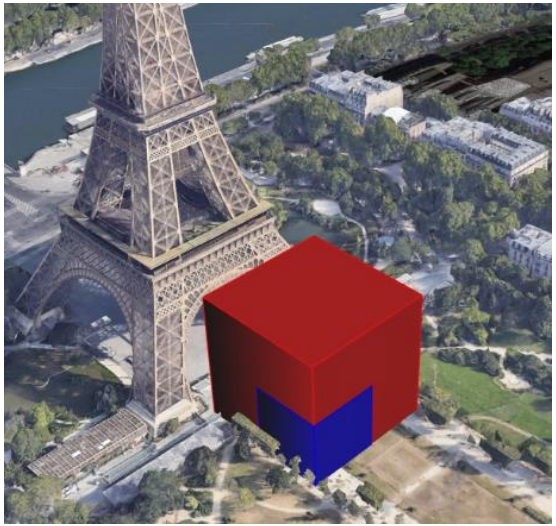
1. Nuclear waste challenges
2. Mont-Terri Experiment
3. Modeling
4. Machine Learning



Nuclear Waste Challenges

Background

Nuclear Waste Around the World



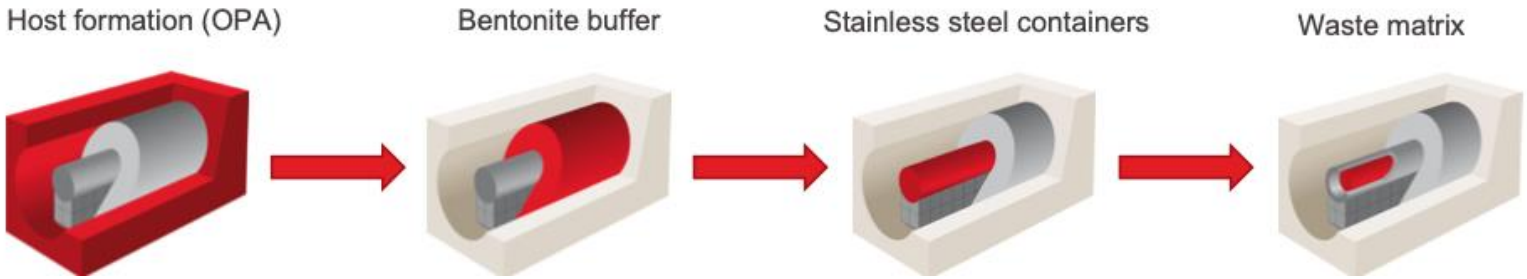
Total volume of High Level Waste in the world (it represents around 400'000 tons so a cube of 64x64x64m; source: world nuclear association, 2019)

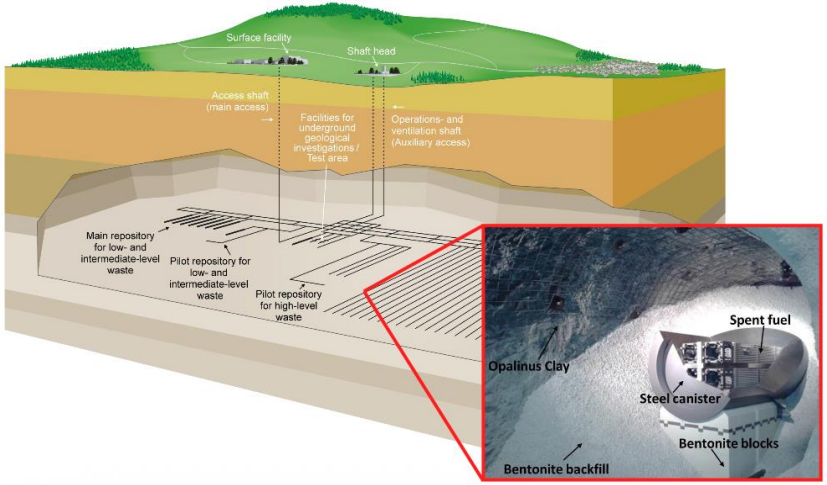
Total Volume of High Level Waste in the EU (it represents around 60'500 tons so a cube of 33x33x33m; source world nuclear association, 2019)

- Radioactive isotopes eventually decay to harmless materials.
- Nuclear wastes isotopes half life range from hours to dozens of thousands of years.
- Need of a proper way to dispose of those wastes

Cesium-137 have half-lives of about 30 years
Plutonium-239 has a half-life of 24,000 years

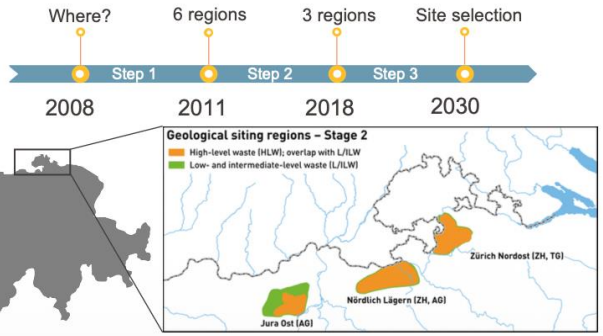
Swiss Repository Concept – Multi-Barrier System





Multibarrier concept

- Vitrified HLW
- Stainless steel container
- Compacted bentonite (granular + blocks)
- Host Clay (Opalinus Clay)



Challenges

Motivation:

- Feasible solution with our current technology
- Protection of waste against **human activities** and **natural events**
- **Relief** of the responsibility for supervision for **future generations**

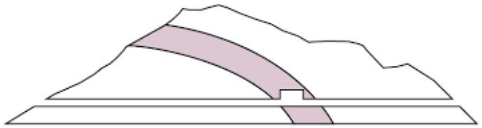
Challenges:

- Heterogeneous materials
- Highly coupled physics
- Lots of material and lots of parameters
- Lack of field data
- Difficult geometry
- Lots of non-linearity
- Long timescale
- Identification of all interactions
- Around ~40 km of gallery should be excavated
- Site selection

Mont-Terri Experiment

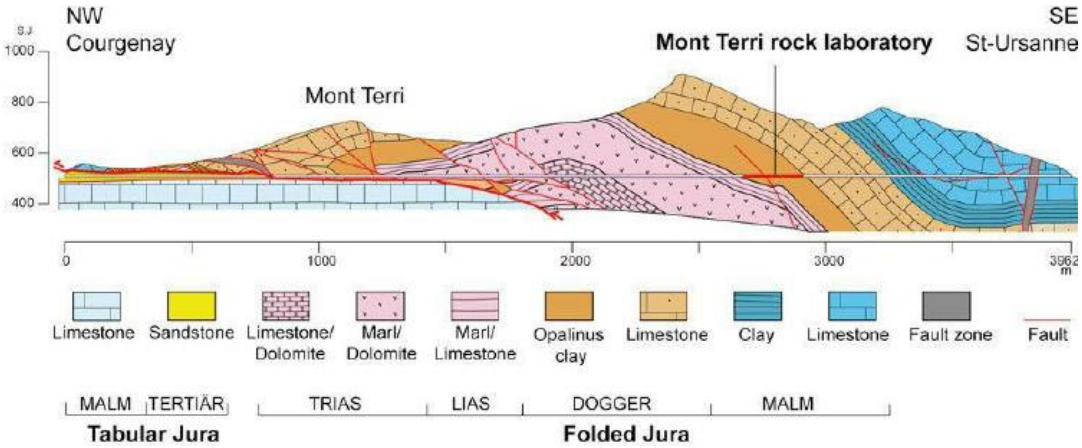
The Mont Terri Project is an international research project for the hydrogeological, geochemical and geotechnical characterisation of a clay formation (Opalinus Clay)

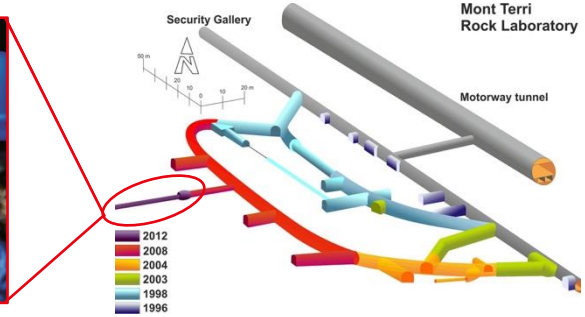
Experiment: goal and location



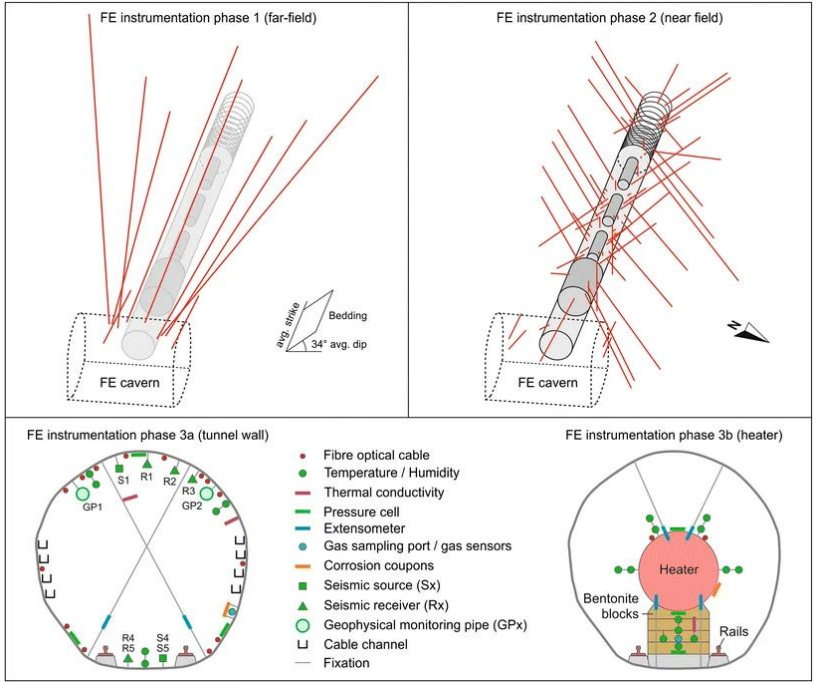
Mont Terri Project

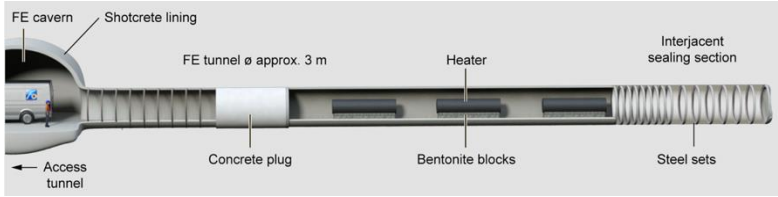
- **Opalinus** clay as host rock and granular bentonite as backfill element.
- It involves the heating of three canisters **under realistic conditions** and the natural hydration
- It is located in the Jura canton around **300 m** deep
- Support: NAGRA, ANDRA, CHEVRON, financial support from the EU for a total of around **10’000’000 €**





Instrumentation of the Mont-Terri Experiment

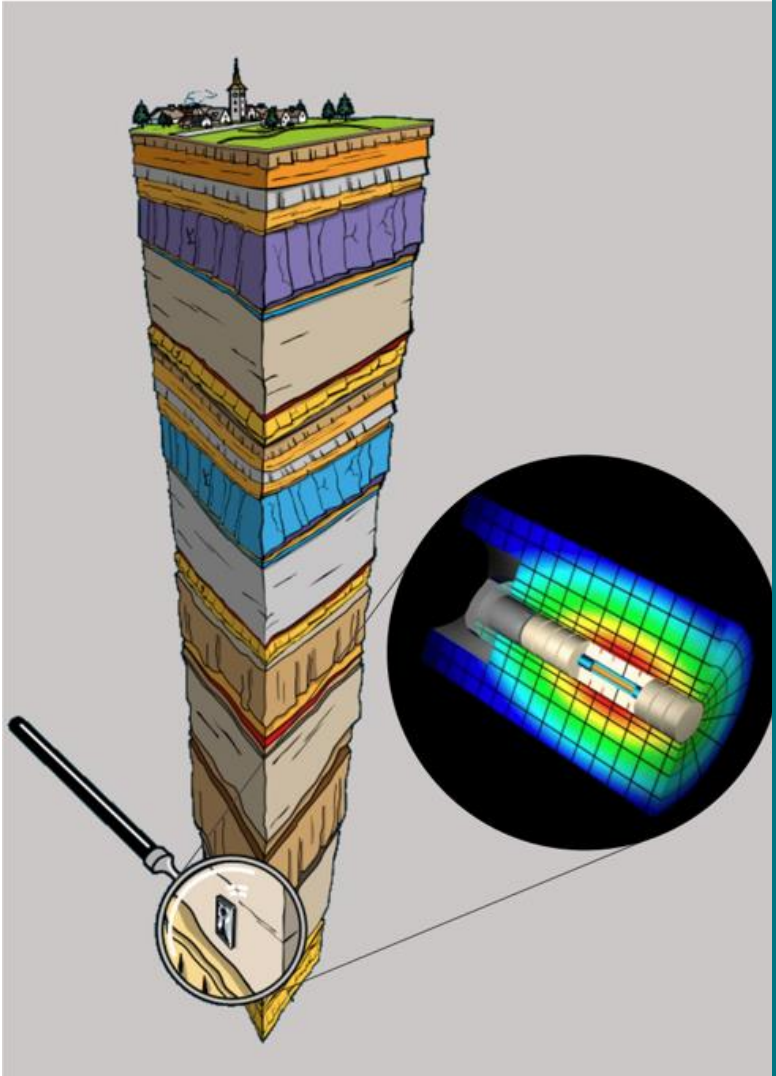




A total of **1’677** sensor were installed including

- 1’200 temperature sensors
- 173 Relative humidity sensors
- 147 Deformation sensors
- 72 Total pressure sensors
- 65 Pore-water pressure sensors
- 20 gas composition

Müller, H.R., Garitte, B., Vogt, T. *et al*. Implementation of the full-scale emplacement (FE) experiment at the Mont Terri rock laboratory. *Swiss J Geosci* **110**, 287–306 (2017). https://doi.org/10.1007/s00015-016-0251-2



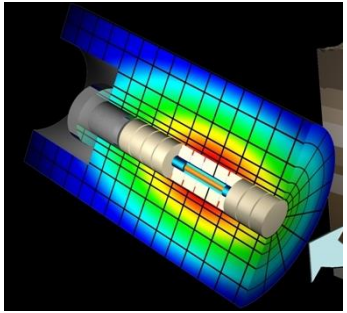
Modeling

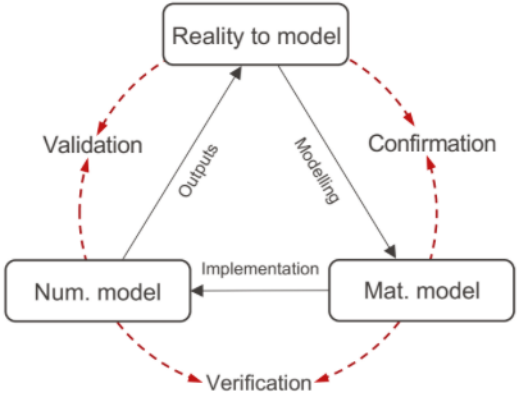
Deterministic approach

Classical Modeling

Why use numerical models?

- Experiments can only give information over a relatively **short timescale**
- To test **different configurations**
- To **forecast the long-term** repository interactions
- Way **cheaper** than lab or in-situ experiments





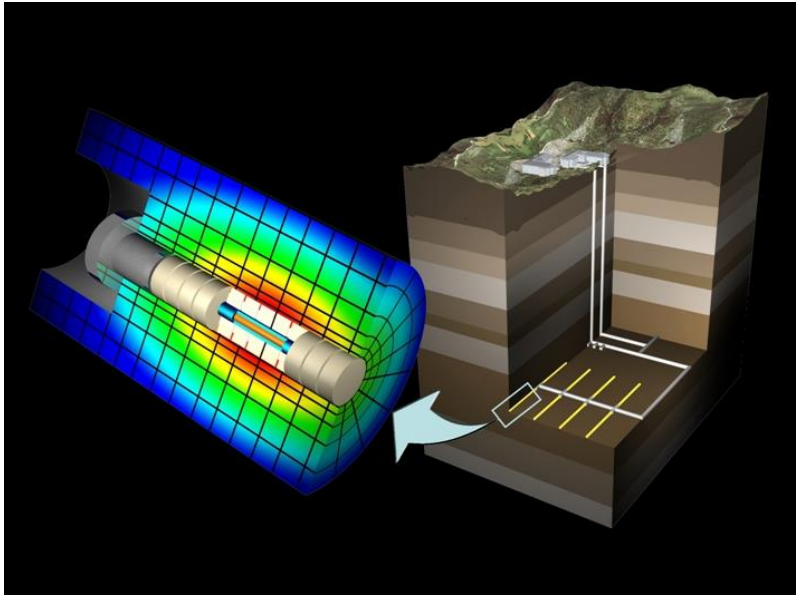
How to proceed?

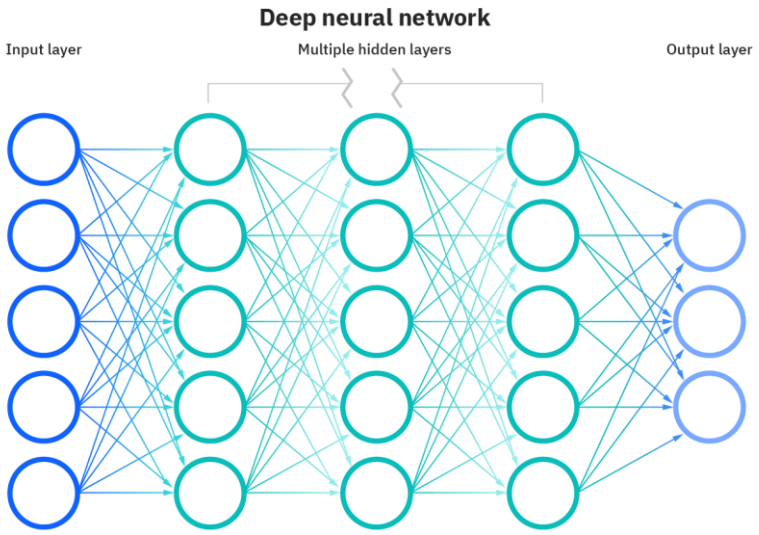
- **Confirm the completeness** of the conceptual **Model** to cover the full spectrum of relevant phenomena and processes.
- **Verify the implementation** in the FEM code.
- **Validate + calibrate the Model**model with the support of the in-situ experiment database

$$\frac{\partial u}{\partial t} = \alpha \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} \right)$$

Model Assumptions and Limitations

- Geometry
- Resolution
- Construction phase
- Identification of the physics
- Empirical models
- Calibration
- Continuum & homogeneous mediums
- Boundary conditions
- Computation demand





Machine Learning

Final Project

Final Project

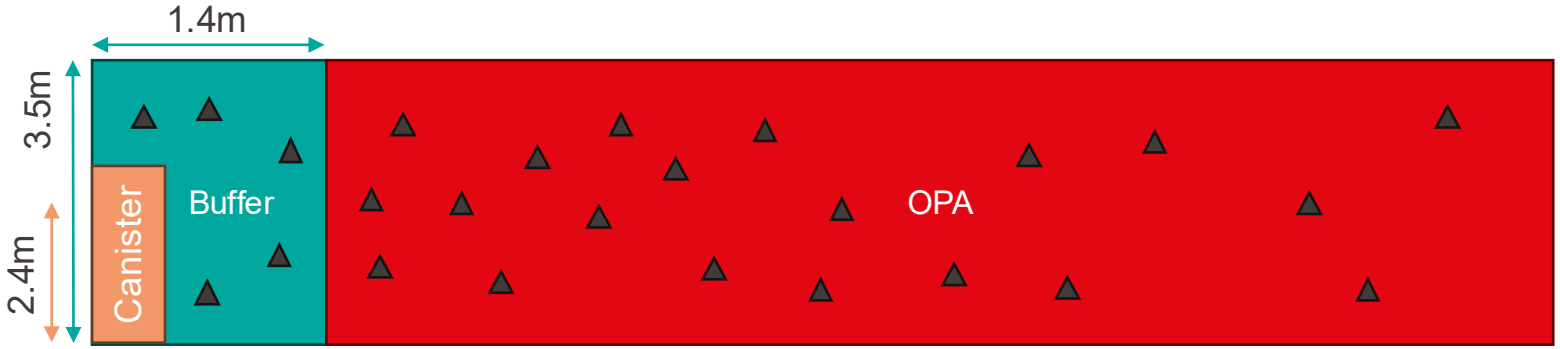


Figure 1: Schematic of the modeled system in 2D (not to scale), including heated canister, buffer and opa. Potential sensor locations are represented as triangles.

Goal

- Use a data-driven method, to predict the temperature of the rocks at locations where we don't have sensors
- Infer temperature at new position across the geometry

Dataset

- Coordinates of sensors and their temperature measurements over time (~250 years)
- Heating power of the canister over the whole duration with three different profiles
- You will receive a train set and will need to evaluate your model against a hidden test set
- The dataset will be provided on moodle
- Be careful with outliers!

Final Project

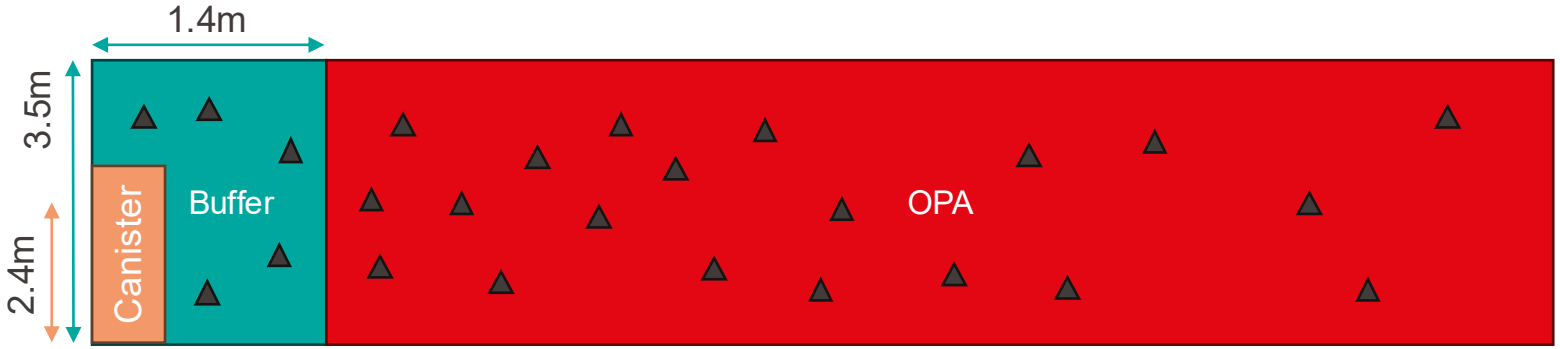


Figure 1: Schematic of the modeled system in 2D (not to scale), including heated canister, buffer and opa. Potential sensor locations are represented as triangles.

- Watch out for
  - Outliers
  - Failed Sensors
  - Sensor drift
- Predictions in OPA are more important in the final metric

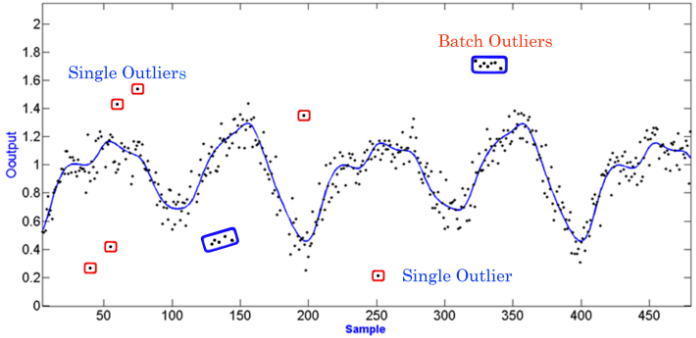


Figure 2: Possible outliers on time series of data not from this project

Logistics

- You submit in teams of 2 (registrations open after the lecture)
- You will need to submit a poster, your code on moodle and your best results on the leaderboard
- There is a leaderboard on Kaggle with a prize for the winner
- Deadline for the Moodle submission (poster+code): 27th of May
- Deadline for the leaderboard: 27th of May
- Grading:
  - 25% poster
  - 25% performance (pass a baseline in Kaggle)
  - 50% methods, results, and code

- Bonus: You will get extra points if you detect sensors that have drift.

Logistics

- Presentation of the Poster is on Thursday, 28th of May in **INJ 218** from 13:00 – 15:00. Bring your Laptop to show your poster.
- It will be like a poster sessions with the Professors and TAs walking around and giving you feedback
- Group registration now open
- Check the notebook on statistical properties of time series (pandas notebook of session 05)

Civil Engineering (GC) / GC - Bachelor / CIVIL-226 / 20 April - 26 April

20 April - 26 April

◀13 April - 19 April ▶27 April - 3 May

Current week

Select your group for the project

Poster examples

- There is an example poster uploaded on Moodle
- It outlines which sections are important in your poster presentation
- Feel free to adapt the example or create a new one